

Recap

$$Y = X\beta$$

$n \times 1 \quad n \times p \quad p \times 1$

$$Y = \begin{pmatrix} y_1 \\ \vdots \\ y_n \end{pmatrix}$$

$$X = \begin{pmatrix} 1 & x_{11} & x_{12} & \dots & x_{1(p-1)} \\ \vdots & \vdots & \vdots & & \vdots \\ 1 & x_{1n} & x_{2n} & \dots & x_{(p-1)n} \end{pmatrix}$$

$$\beta = \begin{pmatrix} \beta_0 \\ \vdots \\ \beta_{p-1} \end{pmatrix}$$

$$\varepsilon = \begin{pmatrix} \varepsilon_1 \\ \vdots \\ \varepsilon_n \end{pmatrix}$$

$$\hat{Y} = X\hat{\beta}$$

$$\hat{\beta} = (X^T X)^{-1} X^T Y$$

$$\hat{Y} = \underbrace{X(X^T X)^{-1} X^T}_H Y$$

$$e = (I - H)Y$$

$$Y \sim N(X\beta, \sigma^2 I_{n \times n})$$

$$\hat{\beta} \sim N(\beta, \sigma^2 (X^T X)^{-1}_{p \times p})$$

$$\text{Var}(\hat{\beta}) = \begin{pmatrix} \text{Var}(\hat{\beta}_0) & & \\ & \ddots & \\ & & \text{Var}(\hat{\beta}_{p-1}) \end{pmatrix}$$

I. Continue w/ Distn's

$$\hat{Y} = X\hat{\beta} = X(X^T X)^{-1} X^T Y = HY$$

$$\hat{Y} \sim N\left(\quad , \quad \right)$$

$$E(\hat{Y}) = E(X\hat{\beta}) = X E(\hat{\beta}) = X\beta$$

$$\text{Var}(\hat{Y})$$

$$\text{Var}(AW) = A \text{Var}(W) A^T$$

$$\text{Var}(\hat{Y}) = \text{Var}(HY)$$

$$= H \text{Var}(Y) H^T$$

$$= H \sigma^2 I H^T$$

$$= \sigma^2 HH^T$$

$$= \sigma^2 H$$

Properties of H :

Any matrix A is a projection matrix if:

(1) A is symmetric

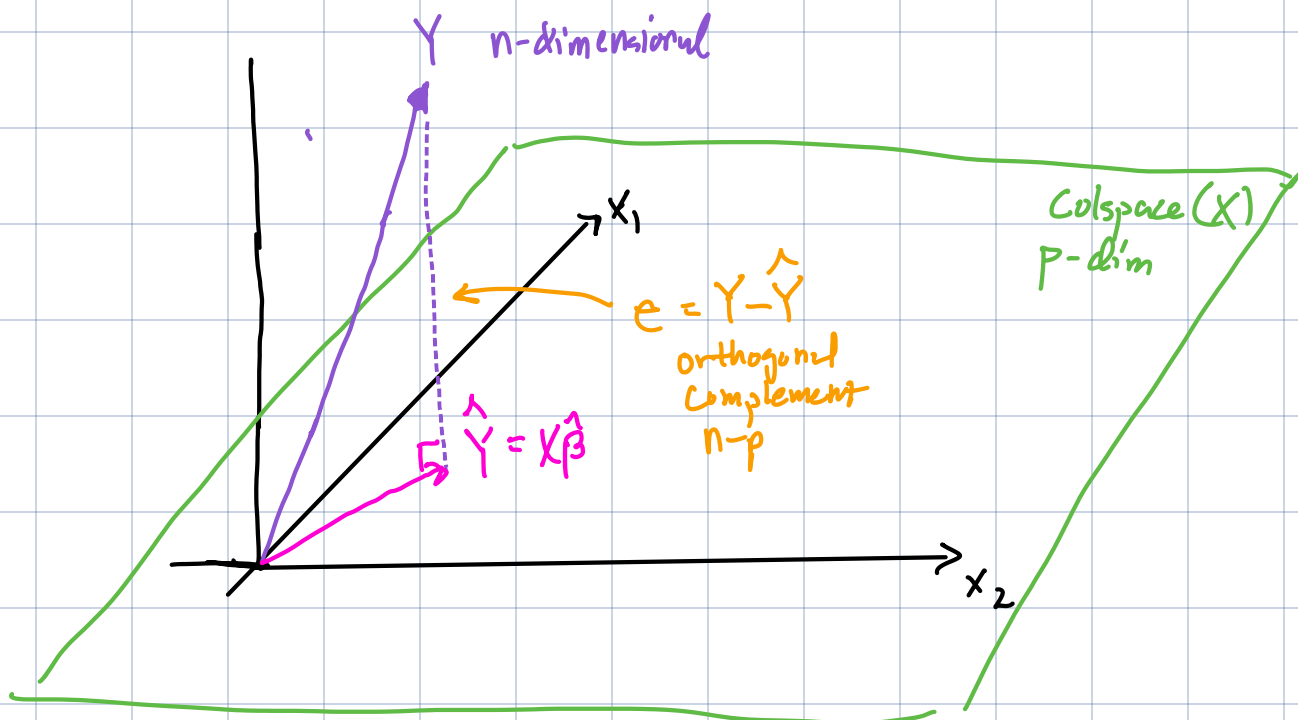
(2) A is idempotent

$$\begin{aligned} \textcircled{1} \quad H^T &= H & (X(X^T X)^{-1} X^T)^T &= X[(X^T X)^{-1}]^T X^T \\ & & &= X[(X^T X)^T]^{-1} X^T \\ & & (X^T X)^T &= (X^T X) \\ & & &= X(X^T X)^{-1} X^T \\ & & &= H \end{aligned}$$

(2) Idempotent $A^2 = A$

$$\begin{aligned} H^2 &= HH = X(X^T X)^{-1} \cancel{X^T X} (X^T X)^{-1} X^T \\ &= X(X^T X)^{-1} X^T \end{aligned}$$

H is a projection matrix!!



$$\sum_i e_i \hat{y}_i = 0 = \langle e, \hat{Y} \rangle$$

$$= e^T \hat{Y}$$

$$= [(I-H)Y]^T HY$$

$$= Y^T (I-H)^T HY$$

$$= Y^T (I-H) HY$$

$$= Y^T (H - H^2) Y$$

$$= Y^T (H - H) Y$$

$$= \emptyset$$

$$e = (I-H)Y$$

$$\hat{Y} = HY$$

e

$$\text{Recall } Y \sim N(X\beta, \sigma^2 I)$$

$$\begin{aligned} e &= Y - \hat{Y} = Y - HY \\ &= (I - H)Y \end{aligned}$$

$$e \sim N(\quad)$$

$$\begin{aligned} E((I - H)Y) &= (I - H)E(Y) \\ &= (I - H)X\beta \\ &= (X - HX)\beta \\ &= (X - X(X^T X)^{-1} X^T X)\beta \\ &= 0 \end{aligned}$$

$$\begin{aligned} \text{Var}((I - H)Y) &= (I - H)\sigma^2 I (I - H)^T \\ &= \sigma^2 (I - H)(I - H) \\ &= \sigma^2 (I - H) \end{aligned}$$

$$e \sim N(0, \sigma^2(I-H))$$

II Estimate σ^2

$$\hat{\sigma}^2 = \frac{SSE}{n-p} = \frac{e^T e}{n-p} = MSE$$

III Global F-test

Are any of my predictors, $X_1, \dots, X_{p-1}$

significant predictors of Y ?

$$H_0: \beta_1 = \beta_2 = \dots = \beta_{p-1} = 0$$

$$H_1: \text{at least one } \beta_j \neq 0$$

$$H_0: y_i = \beta_0 + \varepsilon_i$$

Null/Reduced Model (R)

$$H: y_i = \beta_0 + \beta_1 X_{1i} + \dots + \beta_{p-1} X_{(p-1)i} + \varepsilon_i$$

Full Model (F)

F-statistic:

SSE_R

SSE_F

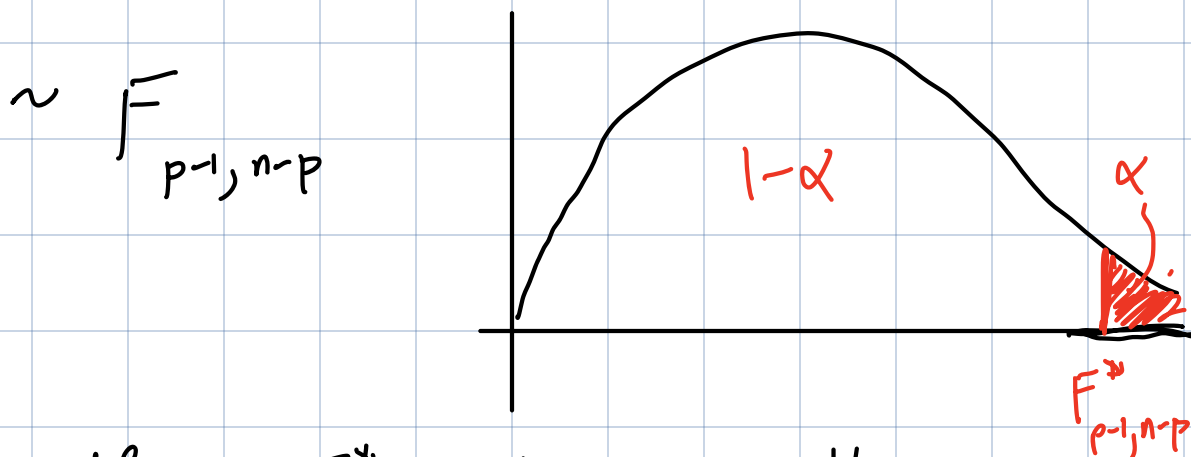
$$(SLR) \quad F = \frac{SSR/1}{SSE/n-2} = \frac{(SST - SSE)/1}{SSE/n-2}$$

(MLR) : Global F-Stat

$$F = \frac{(SSE_R - SSE_F)/p-1}{SSE_F / (n-p)}$$

$p = \# \text{ of } \beta_s$

$$SST = SSG + SSR$$



Decision: If $F > F_{p-1, n-p}^* (1-\alpha)$ reject H_0 .

At least 1 predictor is a significant predictor of Y .

More generically, for any reduced vs full model

$$F = \frac{(SSE_R - SSE_F)/(p-k)}{SSE_F / (n-p)}$$

$K = \# \text{ of } \beta_s \text{ in reduced model}$

ANOVA "Analysis of Variance"

Source	SS	df	MS	F
Regression	$SSE_R - SSE_F$	$p - k$	$SS / p - k$	$\leftarrow 0/B$
Error	SSE_F	$n - p$	$SSE_F / n - p$	$\leftarrow$
Total				

IV Partial F-test

Test a subset of predictors, e.g.

$$X_r, X_{r+1}, \dots, X_{p-1}$$

$$H_0: \beta_r = \beta_{r+1} = \dots = \beta_{p-1} = 0$$

$$H_1: \text{at least one is not } 0$$

$$H_0: y_i = \beta_0 + \beta_1 X_{1i} + \dots + \beta_{r-1} X_{(r-1)i} + \varepsilon_i \leftarrow \text{reduced model}$$

$$H_1: y_i = \beta_0 + \beta_1 X_{1i} + \dots + \beta_{p-1} X_{(p-1)i} + \varepsilon_i \leftarrow \text{full model}$$

$$F = \frac{(SSE_R - SSE_F) / (p - r)}{SSE_F / (n - p)} \quad \leftarrow \text{difference in \# of } \beta\text{'s}$$

$$\sim F_{p-r, n-p}$$

Decision $F > F_{p-r, n-p}^{\alpha}(1-\alpha)$ reject H_0 , at

least one $\{X_r, \dots, X_{p-1}\}$ is a significant predictor of Y conditional on $X_1, \dots, X_{r-1}$ are in the model.